

OMAR HAMMAMI

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ML engineer & interpretability researcher — real-time diffusion steering, representation geometry, GPU systems

INDEPENDENT RESEARCH & PROJECTS

DinoDreamer — Feature Visualization for Frozen Vision Models

2026 – Present

Feature-visualization engine for a frozen DINOv2: gradient-ascent synthesis toward explicit representation-space targets — every run yields an image plus a measured record of what the optimizer actually did.

- Reimplemented core feature-viz methods (MACO, Fourier preconditioning, pyramids, feature accentuation) in one composable engine; targets from Fel et al.'s 32K-atom archetypal SAE or self-trained block-sparse dictionaries
- Every run exactly reproducible: pinned model digests, content-addressed artifact store, replayable RNG/optimizer state; batch stages (dictionary training, synthesis, rendering) run on Modal from source-attested images

SubspaceLens — Information-Theoretic Interpretability for Diffusion Transformers

2026

MCR² coding-rate metrics on a frozen PixArt-a — training-free, no SAEs, no learned probes, just eigenvalues.

- Found progressive subspace separation across PixArt- α 's depth, with $< 1\%$ conditional-vs-unconditional rate difference at the output layer — evidence the organization is architectural, not prompt-driven
- Measured dense activations (Hoyer 0.25–0.71) at the layers doing the most structural reorganization — evidence against the uniform-sparsity assumption SAE methods inherit

Latent Dance (hambajuba2ba) — Real-Time SAE-Steered Diffusion Music Visualizer

2025 – 2026

A live control loop, not batch generation: music re-steers the diffusion model itself every frame — 50 FPS, one consumer GPU.

- Every value in the pipeline is live every frame — audio rewrites the SAE steering vectors (20,480 features, 4 UNet blocks), prompt-embedding interpolation, and noise latents; nothing is a fixed prompt replayed fast
- CUDA graphs replay frozen kernels, so per-frame steering lands via in-place copy/fill into the captured graphs — zero graph breaks, zero recompilation, 50 FPS sustained on a single RTX 5090 with torch.compile
- Auto-labeled all 20,480 SAE features with a 3-VLM ensemble over 50K logged generations — spatial activation heatmaps, TF-IDF fusion; labels published as a HuggingFace dataset
- All audio analysis precomputed (Demucs stems, psychoacoustic features); the render loop never waits on audio
- Desktop app (Electrobun, R3F, GLSL) with an MCP server that lets an LLM agent direct the visuals live

EXPERIENCE

Software Engineering Fellow

Feb 2025 – May 2025

The Recurse Center

New York, NY

- Went deep on systems programming in Rust: built SAM_CAM_BAM, a real-time webcam-segmentation music visualizer (ONNX FastSAM, live frequency-band audio analysis)

Software Engineer (Contract)

Mar 2025 – Apr 2025

The Sentience Company

New York, NY

- Built voice fingerprinting end to end in a 4-week sprint: speaker-embedding API, database schema, SwiftUI onboarding

Machine Learning Engineer

Aug 2023 – Dec 2023

Early-Stage Startup (Stealth)

Washington, DC

- Sole ML engineer: designed and deployed a multimodal computer-vision system (pose detection + CNNs) on GCP
- Rescued a failing pipeline via dataset augmentation and architecture changes, delivering the product's first usable predictions

Machine Learning Student Researcher

Sep 2020 – Jan 2023

NYU C2SMART Center

New York, NY

- Co-authored two peer-reviewed publications on computer vision for smart transportation (below)
- Built vehicle detection and tracking systems (YOLO, Kalman filters); led the dataset curation and annotation team

PUBLICATIONS

- [“A New Curb Lane Monitoring and Illegal Parking Impact Estimation Approach”](#), *Transp. Research Part A* (2022)
- [“Exploring Cost-Effective Computer Vision Solutions for Smart Transportation Systems”](#), USDOT/C2SMART (2023)

EDUCATION

New York University

New York, NY

M.S., Computer Science (2024)

B.S., Computer Science, Minor in Mathematics (2022)

TECHNICAL SKILLS

ML & GPU: PyTorch (torch.compile, CUDA graphs), diffusion models, SAEs & interpretability, ViTs, ONNX

Languages: Python, Rust, TypeScript, C++, SQL, Swift

Infrastructure: Modal, RunPod, Docker, GCP, AWS, Linux, Git, CI/CD, uv

Product: React (R3F), FastAPI, WebSockets, PostgreSQL, GLSL, SwiftUI, Electrobun, MCP